Title: Reproducibility and Geographic Transfer of Satellite-Based Electrification Predictions: Testing High-Resolution Electricity Access Models Across Africa and Asia

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Abstract: Machine learning models that predict electricity access from satellite imagery increasingly guide development investments; over $90 billion in committed funding now flows through portfolios shaped by these tools. The High Resolution Electricity Access (HREA) dataset, which identifies 1.18 billion people in energy poverty across 115 countries, exemplifies this new generation of satellite-derived diagnostics. Yet a core question for operational deployment remains unexamined: can these methods be reproduced, and do they perform consistently across the diverse regions where development banks deploy them?

This study conducts the first independent audit of satellite-based predictions of electrification. Analyzing HREA data from 12 countries across Sub-Saharan Africa and South/Southeast Asia, we test whether estimates can be reproduced using published methods and how accuracy varies between regions where methods were developed and those where they are increasingly applied.

Three findings emerge with direct implications for how multilateral institutions deploy satellite analytics. First, HREA code and outputs are reproducible, confirming baseline transparency standards. Second, HREA correlates strongly with official statistics in Sub-Saharan Africa ($r = 0.92$, $p < 0.01$) but weakly in South Asia ($r = 0.66$, $p = 0.23$), indicating that methods calibrated in low-access African contexts transfer imperfectly to high-access Asian settings. Third, HREA systematically underestimates electrification relative to World Development Indicators, with a mean gap of 18 percentage points across countries.

These findings point to a practical challenge: as development institutions scale satellite-derived tools across regions, regional calibration and validation protocols become essential. We propose a transparency checklist for evaluating satellite-derived energy diagnostics and offer recommendations for strengthening regional calibration before operational adoption. All replication code is released openly.

Keywords: reproducibility; machine learning; electricity access; satellite imagery; nighttime lights; HREA; energy poverty; development finance; South Asia; Sub-Saharan Africa; regional transfer; open science

Data cutoff: All HREA estimates analyzed cover the 2013–2020 period, the full temporal range of the published dataset. World Development Indicator electrification rates were retrieved in October 2025 and reflect the most recent available estimates (2020–2022, depending on country). Policy and investment figures, including World Bank portfolios, Mission 300 commitments, and Nigeria DARES, are current as of November 2025.

Geographic scope: Our analysis covers 12 countries: 6 in Sub-Saharan Africa (Nigeria, Kenya, Ethiopia, Tanzania, Ghana, Uganda) and 6 in South/Southeast Asia (India, Bangladesh, Nepal, Pakistan, Myanmar, Cambodia). Countries were selected based on HREA data availability, variation in electrification levels, and policy relevance to multilateral development bank investments.

Methodological scope: This study employs Tier 1 replication (output verification using published HREA products) rather than Tier 2 replication (re-running the full HREA pipeline from raw VIIRS imagery). Our sensitivity analysis examines temporal stability (2013, 2016, 2020) but does not test alternative settlement layers, background models, or classification thresholds due to computational constraints.

Limitations note: Official electrification statistics from WDI serve as our validation benchmark but are themselves estimates derived from household surveys and administrative records with measurement uncertainty. This study cannot determine whether discrepancies between HREA and WDI reflect HREA measurement error, WDI measurement error, or genuine construct differences between satellite-detected nighttime electricity use and survey-reported household connection status. This distinction matters for operational use: both data sources carry uncertainty, and users should interpret divergence as a signal for further validation rather than definitive error attribution.

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Data Availability Statement: All data used in this analysis are publicly available. HREA settlement-level estimates are accessible via Amazon Web Services (s3://globalnightlight/HREA/). World Development Indicator electrification rates are available from the World Bank Data Catalog (indicator EG.ELC.ACCS.ZS). Administrative boundaries are from GADM version 4.1. Replication code is available at https://github.com/a-novi13-byte/hrea-replication.

1. Introduction

1.1 The Scale of Energy Poverty Reconsidered

Settlement-level electrification estimates now exist for 115 countries, covering 2013-2020, thanks to the High Resolution Electricity Access (HREA) dataset developed by Min et al..[1] Analyzing nighttime satellite imagery across up to 3,000 nights per settlement, HREA identifies 1.18 billion people living in energy poverty, defined as showing no statistical evidence of electricity use at night.[2] This figure stands 60 percent higher than the official World Bank and International Energy Agency estimate of 733 million people lacking electricity access.[3] More troubling still, 447 million people officially counted as electrified show no detectable electricity consumption in the satellite record.[4]

These findings have reshaped how the development community understands the global electrification challenge. The HREA dataset is now distributed through UNDP GeoHub, hosted on Microsoft Planetary Computer, and integrated into the World Bank's Light Every Night repository.[5] It informs the planning tools used by multilateral development banks deploying billions of dollars in infrastructure investment.[6] But a critical operational question remains: how reliable are these estimates across the diverse regional contexts where investment decisions are being made?

1.2 The Policy Stakes

The financial stakes attached to satellite-based electrification measurement are substantial and growing. The World Bank approved $8.1 billion for energy access projects from FY2020 to FY2024.[7] The Energy Sector Management Assistance Program (ESMAP) reports that $46 billion in World Bank development financing has been informed by its analytics through FY2024.[8] The Asian Development Bank invested approximately $12 billion in energy access from 2008 to 2021, providing electricity to over 45 million households.[9] The African Development Bank committed $12.74 billion for energy investments from 2016 to 2025 through its New Deal on Energy for Africa initiative.[10]

These investments are accelerating. Mission 300, launched jointly by the World Bank and African Development Bank in April 2024, targets 300 million new connections in Sub-Saharan Africa by 2030 with a funding goal of $90 billion. Partners have already pledged over $50 billion, with the World Bank contributing $30 billion in IDA resources.[11] Individual operations have reached unprecedented scale: Nigeria's Distributed Access through Renewable Energy Scale-up (DARES), approved in December 2023, represents the World Bank's largest single distributed energy project globally, with total program financing exceeding $2 billion and a target of reaching 17.5 million Nigerians.[12]

Satellite-based tools are increasingly embedded in the planning infrastructure supporting these investments. ESMAP's Global Electrification Platform covers 58 countries with 96 planning scenarios, while the World Bank's Light Every Night repository provides approximately 250 terabytes of nighttime satellite imagery spanning 1992 to 2020.[13] HREA products are housed on this same infrastructure, offering settlement-level estimates that inform geospatial targeting of investments.[14] Yet the transition from research dataset to operational planning tool has proceeded

largely without systematic validation across the diverse regional contexts where these diagnostics are now deployed. This gap matters: tools that perform well in one setting may introduce systematic bias when applied elsewhere, with direct consequences for investment targeting and resource allocation.

The regional diversity of electrification contexts compounds this challenge. Sub-Saharan Africa averages 47 percent electrification, with 565 million people lacking access, representing 85 percent of the global deficit. Within the region, rates range from 5.4 percent in South Sudan to 87.7 percent in South Africa, an 82.3 percentage-point span. South Asia, by contrast, has achieved approximately 97 percent average access, with India and Bangladesh both at 99.5 percent. Southeast Asia exceeds 95 percent on average, though Myanmar lags at 76.8 percent.[15] Methods calibrated in one context cannot simply be assumed to transfer reliably to another, and as development portfolios expand across regions, the need for rigorous cross-regional validation becomes operationally urgent.

1.3 The Reproducibility Gap

The HREA methodology was developed and validated primarily in African contexts. The foundational work by Min et al. demonstrated the relationship between nighttime lights and electrification using data from Senegal and Mali.[16] Subsequent development and calibration drew heavily on household survey data from Sub-Saharan African countries, where electrification rates remain low, and settlement patterns are distinct. The 50-percentage-point baseline difference between Sub-Saharan Africa's 47 percent average access and South Asia's 97 percent average represents a fundamentally different measurement environment.[17] This raises a core operational question: do methods calibrated in low-access African settings transfer reliably to high-access Asian contexts where development banks are increasingly deploying them?

The regional transfer of satellite-based electrification models is non-trivial for several reasons. Settlement patterns differ markedly: the dense, nucleated villages common in South Asian river valleys produce nighttime light signatures distinct from the dispersed homesteads typical of much of rural Africa. Technology mixes vary as well; 55 percent of new connections in Sub-Saharan Africa from 2020 to 2022 came from off-grid solar systems, which produce weaker, more variable nightlight signatures than the grid connections dominant in Asia.[18] Infrastructure configurations, including the prevalence of street lighting versus household-only electrification, further complicate cross-regional comparisons.

Documented limitations of nighttime lights compound these concerns. Detection thresholds leave dim lighting, off-grid solar home systems, and LED lights below satellite sensitivity.[19] Light blooming extends visible radiance approximately 3.5 kilometers beyond actual illuminated areas.[20] Villages with unreliable power show reduced correlation between detectable lights and actual electrification status. Productive uses of electricity and indoor-only consumption remain undetected. Villages without public streetlights may appear dark even when homes are electrified.

HREA represents a significant methodological advance: the VIIRS Day-Night Band offers 742-meter resolution compared to the 2.7-kilometer resolution of its DMSP-OLS predecessor, with a dynamic range seven orders of magnitude wider and on-board calibration that DMSP-OLS

lacked.[21] The research team has published code and technical documentation, meeting baseline standards for reproducibility.[22] Yet publication does not equal verification. To our knowledge, no independent research team has attempted to reproduce HREA estimates, tested the methodology outside its original African calibration context, or quantified the sensitivity of outputs to specification choices. As HREA becomes embedded in operational tools directing billions in investment, this validation gap carries growing consequences for how resources are targeted and allocated.

1.4 Research Questions

This paper addresses five questions central to the responsible deployment of satellite-based electrification diagnostics in development operations:

Table 1 – Research Questions

RQ1	Can HREA estimates be reproduced using published code and data?
RQ2	Do estimates show consistent accuracy across Sub-Saharan Africa and South/Southeast Asia?
RQ3	How sensitive are estimates to specification choices, and does sensitivity vary by region?
RQ4	How well do HREA estimates correlate with official statistics across regions?
RQ5	What transparency standards should multilateral development banks apply when deploying These tools in operational settings?

1.5 Contributions

This study makes five contributions to the evidence base on satellite-based electrification measurement and its operational application.

First, we provide the first independent reproducibility audit of HREA, testing whether published code and data produce stated estimates when executed by researchers outside the original team. Second, we offer the first systematic cross-regional comparison of HREA performance, contrasting accuracy in Sub-Saharan Africa, where the methodology was developed, with South and Southeast Asia, where electrification contexts differ substantially. Third, we quantify specification sensitivity by region, examining how estimate stability varies with parameter choices across different measurement environments. Fourth, we develop a practical transparency checklist for multilateral development bank evaluation of satellite-derived diagnostics, translating methodological findings into actionable guidance for procurement and deployment decisions. Fifth, we release complete replication code openly, enabling other researchers and operational teams to verify findings and extend the analysis to additional regions or time periods.

Together, these contributions aim to strengthen the evidentiary basis for satellite-based tools now embedded in decisions affecting hundreds of millions of people and tens of billions of dollars in development investment. Our goal is not to challenge the value of satellite-derived electrification diagnostics, which represent a significant advance in measurement capability, but to support their responsible deployment by identifying where regional calibration and validation protocols can improve accuracy and reliability.

2. Background and Literature

2.1 Satellite-Based Electrification Prediction

The scientific foundation for satellite-based electrification measurement emerged from the convergence of advances in remote sensing and development economics in the early 2010s. Henderson, Storeygard, and Weil (2012) established the foundational framework in the *American Economic Review*, demonstrating that a 1-percentage-point change in nighttime luminosity corresponds to approximately a 0.28–0.30 percentage-point change in GDP growth.[23] For countries with unreliable national accounts, their analysis showed that optimal growth estimates should weight nighttime lights data roughly equally with conventional economic measures. This work laid the statistical foundation for using satellite imagery as a development diagnostic—and signaled the potential for remote sensing to fill critical data gaps where traditional measurement infrastructure is weak.

Min et al. (2013) provided the first systematic ground-based validation of nighttime lights for detecting rural electrification. Working with survey data from 232 electrified and 899 unelectrified villages in Senegal and Mali, they found that electrified settlements appear brighter primarily because of streetlights rather than household electricity use. Their threshold calibration methodology demonstrated that DMSP-OLS monthly composites record a one-point brightness increase on the instrument's 63-point scale for every 60 to 70 additional streetlights or 240 to 270 electrified homes.[24] This finding carries important implications for interpreting satellite-based estimates in contexts where streetlighting prevalence varies, a factor that differs substantially between African and Asian electrification patterns.

The transition from DMSP-OLS to VIIRS instrumentation transformed measurement capabilities. Elvidge et al. (2013, 2017) documented that the VIIRS Day-Night Band provides 742-meter resolution compared to DMSP-OLS's approximately 2.7-kilometer resolution, with a 14-bit dynamic range versus 6-bit, and on-board calibration that enables consistent radiometric measurements across time.[25] These technical improvements eliminated saturation in bright urban cores and enhanced detection of dim rural settlements, making high-resolution electrification tracking operationally feasible for the first time.

Machine learning approaches extended this foundation in two directions. Jean et al. (2016) demonstrated in *Science* that convolutional neural networks trained on daytime imagery using transfer learning could explain up to 75 percent of the variation in local-level economic outcomes across five African countries, pioneering the combination of nighttime lights as intermediate labels with daytime imagery features.[26] Falchetta et al. (2019) created the first fully reproducible high-resolution gridded electrification dataset for Sub-Saharan Africa at 1-

kilometer resolution, achieving $R^2 = 0.81$ against both national-level Tracking SDG 7 data and province-level DHS survey data while integrating the World Bank Multi-Tier Framework for measuring electricity consumption quality.[27] These advances demonstrated that satellite-derived diagnostics could achieve accuracy sufficient for policy-relevant applications.

The HREA methodology represents a fundamental departure from threshold-based classification. Rather than classifying settlements as electrified when brightness exceeds fixed cutoffs, Min et al. (2024) employ statistical anomaly detection comparing observed brightness against expected background noise.[28] The methodology trains a linear mixed model on isolated non-settlement cells at least 15 arcseconds from populated areas, generating expected brightness values for every location on every night. Settlements are classified as electrified when brighter than anticipated, producing an Artificial Light Score ranging from 0 to 1, representing the probability of electrification. This probabilistic approach offers advantages for identifying areas with unreliable or intermittent supply that binary classification misses, a potentially significant gain for targeting investments in contexts where grid reliability varies. Yet the methodology was developed and calibrated primarily in African contexts, raising the question of whether its performance generalizes to the high-access Asian settings where development institutions increasingly deploy it.

Table 2 – Evolution of Satellite-Based Electrification Methods

STUDY	YEAR	METHOD	KEY INNOVATION	PRIMARY VALIDATION REGION
Henderson et al.	2012	Luminosity-GDP correlation	Economic activity proxy framework	Global (cross-country)
Min et al.	2013	Threshold calibration	First ground-validated electrification detection	Senegal, Mali
Elvidge et al.	2013/2017	VIIRS instrumentation	742m resolution, on-board calibration	Global
Jean et al.	2016	CNN transfer learning	Daytime imagery + nightlight labels	Five African countries
Falchetta et al.	2019	Random forest + multi-tier	Reproducible 1km gridded dataset, $R^2 = 0.81$	Sub-Saharan Africa
Min et al. (HREA)	2024	Statistical anomaly detection	Probabilistic scores, nightly analysis	115 developing countries

Operational deployment has moved these methods from research to policy application, and the scale of adoption is substantial. The Global Electrification Platform covers 58 countries and 96 planning scenarios integrating grid, mini-grid, and off-grid solutions.[29] The World Bank's Light Every Night repository provides approximately 250 terabytes of nighttime satellite imagery spanning 1992 to 2020, hosting HREA products alongside raw imagery.[30] UNDP GeoHub's Electricity Access Dashboard offers visualization for over 100 countries with forecasting through 2030, while the WRI Energy Access Explorer combines supply and demand indicators for multi-criteria analysis across 15 countries, serving more than 25,000 users, including over 100 governments.[31] This infrastructure now underpins investment targeting across multilateral portfolios, which makes the question of cross-regional validity not merely academic but operationally consequential.

2.2 The Reproducibility Challenge in Applied ML

Economics research faces persistent reproducibility challenges that are well documented but incompletely addressed, a concern that extends directly to the econometric and machine learning tools increasingly embedded in development operations. Christensen and Miguel (2018) provide the definitive survey in the *Journal of Economic Literature*, concluding that publication bias, inability to replicate, and specification searching remain widespread.[32] Camerer et al. (2016) replicated 18 experimental economics studies from top journals, finding a 61 percent replication rate with replicated effect sizes averaging only 66 percent of original estimates.[33] Without author assistance, Chang and Li (2015) found that only approximately 33 percent of economics papers could be replicated; with author assistance, this rate increased to only 50 percent.[34] These findings raise a practical question for development institutions: when research outputs are translated into operational tools, what verification standards should apply?

Machine learning research faces parallel challenges with distinct characteristics. Kapoor and Narayanan (2023) documented data leakage across 294 papers spanning 17 scientific fields, creating a taxonomy of 8 leakage types ranging from textbook errors to open problems.[35] Their analysis of civil war prediction models showed that after correcting for leakage, complex machine learning models perform no better than decades-old logistic regression, a sobering finding for institutions investing in ML-based diagnostics. The NeurIPS 2019 Reproducibility Program found that while 97 percent of submissions claimed clear model descriptions, verified code availability reached only 27.7 percent when reviewers checked directly rather than relying on author self-reports.[36] The gap between claimed and verified reproducibility suggests that standard academic transparency practices may be insufficient for tools transitioning to operational use.

Random seed sensitivity compounds these issues. Analysis of 85 ACL Anthology publications found that over 50% contain risky uses of random seeds, and some natural language processing tasks show variance exceeding 18 percentage points due to seed choice alone.[37] For satellite-based applications specifically, researchers' degrees of freedom multiply: cloud masking algorithms vary in accuracy from 80 to 97 percent depending on implementation, compositing periods affect signal stability, and gap-filling strategies for cloudy observations introduce additional variation.[38] These specification choices are rarely documented in ways that enable operational users to assess their implications for estimate reliability.

HREA represents a critical test case for satellite-based reproducibility in development applications. The research team has published full replication code on GitHub under the GPL-3.0 license, with Cloud-Optimized GeoTIFFs accessible via AWS, meeting baseline transparency standards that many datasets do not. However, documentation explicitly notes that processing is not recommended for personal machines and requires supercomputing clusters, creating barriers to independent verification.[39] No independent research team has attempted to reproduce HREA estimates, tested the methodology outside its original calibration context, or quantified sensitivity to specification choices. As HREA products become embedded in operational tools directing billions in investment, this verification gap carries growing consequences. The question is not whether HREA represents a methodological advance, it does, but whether the transition from

research output to policy instrument has been accompanied by the validation protocols that operational deployment requires.

2.3 Regional Contexts: Africa vs. South/Southeast Asia

Sub-Saharan Africa accounts for 85 percent of the global electricity access deficit, up from 50 percent in 2010, despite absolute numbers declining slightly (Tracking SDG 7 2025).[40] Regional access averages 47 percent, with extreme variation ranging from 5.4 percent in South Sudan to 87.7 percent in South Africa, an 82.3 percentage-point span.[41] Kenya represents a notable success, doubling its electrification rate from 36 percent in 2014 to 76.2 percent in 2023 through a combination of grid and off-grid strategies.[42] Ethiopia maintains only 55.4 percent access despite hosting the continent's third-largest access deficit at 56.4 million people, with stark urban-rural divides: 96 percent urban versus 27 percent rural and only 5 percent in deep rural areas.[43] This heterogeneity within Sub-Saharan Africa; in access levels, technology pathways, and settlement patterns, means that even methods calibrated for the region may perform unevenly across countries.

The technology mix in Sub-Saharan Africa differs markedly from other regions, with direct implications for satellite-based measurement. The ESMAP Off-Grid Solar Market Trends Report 2024 documents that 55 percent of new electricity connections in Sub-Saharan Africa between 2020 and 2022 came through off-grid solar systems rather than grid extension.[44] Small solar home systems and pico-solar products producing approximately 100 lumens routinely generate lighting levels below the VIIRS background classification threshold of approximately 0.6 $nW/cm^2/sr$.[45] This creates a systematic underestimation of electrification from dim sources, a measurement challenge that threshold-based methods cannot efficiently address, and probabilistic approaches like HREA aim to mitigate.

South and Southeast Asia present fundamentally different electrification landscapes. India achieved 99.5 percent access through the Saubhagya program, which electrified 28.6 million households between 2017 and 2022.[46] Bangladesh reached the same milestone in 2023, progressing from 62 percent in 2010.[47] Nepal reports 94 percent access, while Pakistan nominally shows 95.6 percent, though severe reliability issues mean only 3 percent of households receive truly reliable supply, with 79 percent receiving power for only a few hours daily.[48] Vietnam maintains 99.8 percent access, and Cambodia has achieved rapid gains, reaching approximately 95 percent, with power generation increasing 14-fold since 2010.[49] Myanmar remains the regional outlier at 76.8 percent of households with access, with only approximately 20 percent of rural areas connected to the grid.[50] For satellite-based diagnostics, this context presents a different measurement challenge: the task shifts from detecting the presence of electrification to assessing the quality and reliability of supply, a distinction that nighttime light signatures may not fully capture.

These regional differences have direct implications for model transferability. Dugoua, Kennedy, and Urpelainen (2018) validated nighttime lights against the 2011 Census of India, finding that accuracy degrades substantially when power supply is intermittent.[51] In states with high electrification and reliable supply, such as Punjab and Tamil Nadu, correlations between satellite estimates and ground truth are high; in states with low electrification and intermittent supply,

such as Bihar and Uttar Pradesh, correlations are substantially lower. Settlement patterns also differ markedly: the dense, nucleated villages typical of South Asian river valleys produce nighttime light signatures distinct from the dispersed homesteads common in much of rural Africa. Methods validated in contexts where many settlements remain entirely unelectrified, where the distinction between lit and unlit is relatively straightforward, may perform differently in contexts where the challenge shifts to distinguishing reliable from unreliable supply. This is the core transferability question our analysis addresses: does a methodology calibrated to detect the presence of electricity in low-access African settings maintain accuracy when deployed in high-access Asian contexts where the policy-relevant margin has shifted to supply quality?

2.4 Hypotheses

Building on the evidence reviewed above, we test four hypotheses concerning HREA reproducibility and regional transfer:

Table 3 – Hypotheses

H1 (Reproducibility)	HREA estimates can be reproduced within reasonable tolerance using published code and openly available data.
H2 (Regional variation)	HREA prediction accuracy will differ between Sub-Saharan Africa and South/Southeast Asia, with stronger performance in African contexts where methods were calibrated.
H3 (Discrimination capacity)	HREA will show greater ability to distinguish electrified from unelectrified settlements in low-access African contexts than in near-universally electrified Asian settings.
H4 (Sensitivity patterns)	Specification sensitivity will be greater in regions further from HREA's primary calibration context.

These hypotheses move from baseline verification (can estimates be reproduced?) to operational validity (do they perform consistently where development institutions deploy them?). Section 3 describes the data and methods used to test these hypotheses across 11 countries in Sub-Saharan Africa and South/Southeast Asia.

3. Data and Methods

3.1 Data Sources Overview

Our analysis draws on four openly accessible data sources requiring no registration or institutional credentials, a design choice that reflects both scientific best practice and the practical realities of development research. The HREA dataset provides settlement-level

electrification estimates for 115 countries, distributed as Cloud-Optimized GeoTIFFs via the World Bank's Light Every Night repository on Amazon Web Services. The dataset spans 2013 to 2020, offering annual estimates for each settlement based on analysis of thousands of nightly satellite observations. In our download of the 12 countries examined in this study, the HREA data total approximately 34 gigabytes, with India alone accounting for 16.1 gigabytes due to its subdivision into 37 state-level files.[52]

Official electrification rates come from the World Bank's World Development Indicators, specifically indicator EG.ELC.ACCS.ZS measuring the percentage of the population with access to electricity. These nationally representative statistics derive from household surveys and administrative records, providing the validation benchmark against which satellite estimates are evaluated. Administrative boundaries from the Global Administrative Areas Database (GADM) version 4.1 enable aggregating settlement-level HREA estimates to subnational units, facilitating validation when provincial or state-level electrification data are available.[53] The HREA methodology is documented in a public GitHub repository that contains the full processing pipeline, from raw satellite imagery to final electrification estimates.

Table 4 – Data Sources

DATA TYPE	SOURCE	RESOLUTION	ACCESS METHOD	VOLUME
HREA electrification estimates	AWS Light Every Night	Settlement-level (~1 arcsec)	aws s3 cp --no-sign-request	34 GB
HREA replication code	GitHub	N/A	Git clone	<1 MB
Official electrification rates	World Bank WDI	National	Direct download	<5 MB
Administrative boundaries	GADM v4.1	Vector (Admin-1)	Direct download	~500 MB

Our commitment to fully open data reflects a principle central to responsible deployment of satellite-derived diagnostics: validation should be accessible to all users, not only those with institutional resources. Development practitioners and researchers in low-resource settings cannot access datasets requiring expensive subscriptions or institutional affiliations. Every data source used in this analysis can be obtained by any researcher with internet access and standard computing equipment. This accessibility matters operationally; if the tools informing investment decisions cannot be independently verified by the institutions and governments using them, the evidentiary basis for those decisions remains incomplete.

3.2 Country Selection

Our sample comprises 12 countries selected to maximize variation in electrification contexts while maintaining comparability across regions. Six countries represent Sub-Saharan Africa: Nigeria, Kenya, Ethiopia, Tanzania, Ghana, and Uganda. Five countries represent South Asia: India, Bangladesh, Nepal, Pakistan, and Myanmar. Cambodia provides Southeast Asian representation. This distribution enables systematic comparison of HREA performance across

the regional contexts where satellite-based electrification diagnostics are most actively deployed in development operations.

Selection criteria prioritized four dimensions. First, HREA coverage: all selected countries have complete settlement-level estimates for 2013 to 2020. Second, variation in electrification rates: official 2020 access rates range from 39.9 percent in Tanzania to 96.5 percent in India, spanning the full range from early-stage electrification to near-universal access. Third, regional representation: the sample includes both Sub-Saharan African contexts, where HREA methodology was primarily developed, and South Asian contexts, where distinct infrastructure configurations and settlement patterns may affect measurement validity. Fourth, policy relevance: all 12 countries receive substantial multilateral development bank investment in electrification, making the accuracy of satellite-based targeting tools directly consequential for resource allocation. Together, these criteria ensure that our findings speak to the contexts where validation matters most: not edge cases, but the core of development portfolios.

Table 5 – Selected Countries

Country	Region	WDI 2020 (%)	Settlements	Admin-1 Units	Selection rationale
Nigeria	SSA	55.4	17.4M	37	Largest African population
Kenya	SSA	71.5	5.5M	47	Rapid recent progress
Ethiopia	SSA	51.1	8.5M	11	Large, lower-access baseline
Tanzania	SSA	39.9	11.9M	31	Lowest access in sample
Ghana	SSA	85.4	3.9M	16	High-access African comparator
Uganda	SSA	42.1	4.7M	58	East African comparison
India	SA	96.5	84.0M	41	Near universal, largest settlement count
Bangladesh	SA	96.2	4.2M	8	Rapid progress, high density
Nepal	SA	89.9	3.0M	5	Geographic challenges
Pakistan	SA	94.5	17.0M	8	Large South Asian comparator
Myanmar	SEA	70.4	0.4M	15	Lower-access Asian case
Cambodia	SEA	86.4	2.4M	25	Southeast Asian representation

The sample encompasses 162.8 million settlements across 302 administrative units at the first subnational level. India dominates the settlement count, contributing 84.0 million of the total, reflecting both its population scale and the relatively fine-grained settlement layer underlying HREA estimates in South Asia. The African countries collectively contribute 52.0 million settlements, enabling robust cross-regional comparisons despite differences in absolute scale. This coverage provides sufficient statistical power to detect meaningful regional differences in HREA performance; the central question for operational deployment.

3.3 HREA Data and Methodology

3.3.1 HREA Outputs

The HREA dataset provides four variables for each settlement-year combination, each capturing a distinct dimension of electrification status relevant to operational targeting. The primary outcome measure is set_lightscore, which represents the estimated probability that a settlement is electrified and is scaled from 0 to 1. This probabilistic measure captures uncertainty in electrification status rather than forcing binary classification, offering advantages for identifying areas with unreliable or intermittent supply, a distinction that matters for differentiating investment needs.

Three additional variables provide complementary information. The set_zscore variable reports the standardized residual comparing observed settlement brightness to expected background brightness, expressed in standard deviation units; positive values indicate settlements brighter than predicted for their location and characteristics. The set_prplit variable measures the proportion of observed nights on which the settlement appeared statistically brighter than its matched background, ranging from 0 to 1. This measure captures supply reliability: a settlement with high lightscore but low proportion of lit nights may have access but intermittent supply, precisely the quality dimension that conventional binary electrification statistics miss. The rade9lnmu variable provides the annual composite of nightlight radiance in log-transformed mean units, enabling comparison to other nighttime lights research using radiance-based measures.

All four variables are distributed as Cloud Optimized GeoTIFFs at approximately 1 arcsecond resolution in the WGS84 coordinate reference system (EPSG:4326). Missing values are coded as -999999 and stored in float32 format. The cloud-optimized format enables efficient partial reads, allowing researchers and operational teams to extract data for specific regions without downloading entire country files, a practical advantage for institutions working with bandwidth or storage constraints.

3.3.2 HREA Methodology

The HREA methodology represents a fundamental departure from threshold-based electrification classification, and understanding this design is essential for interpreting both its strengths and its potential limitations across contexts. Rather than classifying settlements as electrified when nightlight brightness exceeds fixed cutoffs, the approach employs statistical anomaly detection comparing each settlement's observed brightness to expected background brightness for its location.

The core innovation is a linear mixed model trained on isolated non-settlement cells at least 15 arcseconds from any populated area. This model learns the expected brightness pattern for uninhabited terrain, accounting for factors such as moonlight, atmospheric conditions, and instrument characteristics. For each settlement on each night, the model generates a predicted

brightness value representing what the location would show if unelectrified. The background model is calibrated regionally, which raises the question we examine empirically: does calibration in one regional context transfer reliably to another?

Settlements are classified as statistically brighter than expected when observed radiance exceeds the model prediction by a threshold determined from the background distribution. The lightscore aggregates these nightly classifications across all available observations, typically spanning 2,500 to 3,000 nights depending on cloud cover patterns. A settlement appearing brighter than expected on 80 percent of clear nights receives a lightscore of approximately 0.8, indicating high probability of electrification with reliable supply. This temporal aggregation provides robustness against transient anomalies—a methodological strength—but also means that systematic regional differences in background brightness or settlement characteristics could propagate through the entire time series.

This probabilistic approach offers advantages over binary classification for development applications. Areas with intermittent supply, common where grid infrastructure is overstretched, show intermediate lightscore values rather than being forced into electrified or unelectrified categories. The methodology also naturally accounts for the substantial night-to-night variation in satellite observations, reducing sensitivity to outliers from transient light sources or atmospheric anomalies. For operational targeting, these features represent genuine advances: they enable planners to distinguish between areas with no access and those with access but poor reliability, different problems requiring different interventions.

3.3.3 Replication Approach

The HREA research team has released full processing code through a public GitHub repository under the GPL-3.0 license, a transparency standard that enables independent verification. The repository contains seven core functions executed in sequence: boundary preparation, population grid creation, settlement identification, VIIRS data acquisition and cleaning, model fitting, metric computation, and output file generation. Dependencies include the R packages data.table, terra, geodata, MODISTools, and lme4 for mixed model estimation.

We adopt a tiered replication approach that explicitly reflects computational constraints acknowledged in the HREA documentation. The repository README notes that full replication is not recommended for personal computing equipment due to substantial disk, RAM, and CPU requirements; cluster computing resources are advised. This constraint is itself operationally relevant: if independent verification requires access to supercomputing, most institutions deploying HREA products cannot verify the estimates they rely on. Given this reality, our analysis proceeds in three tiers.

Tier 1 (Output Verification) compares downloaded HREA Cloud Optimized GeoTIFFs to statistics published in Min et al. (2024), confirming that distributed data products match documented findings. Tier 2 (Code Inspection) examines published code for reproducibility

barriers, including hardcoded paths, missing dependencies, and undocumented parameter choices. Tier 3 (Partial Replication) tests specific pipeline components where computationally feasible, focusing on the statistical model specification and aggregation procedures rather than full end-to-end processing.

This tiered approach enables meaningful reproducibility assessment within practical resource constraints while transparently documenting the boundaries of verification. We are explicit about what we can and cannot independently verify, a practice we recommend for any audit of satellite-derived diagnostics. The distinction matters for operational users: where verification is incomplete, confidence in estimates should be calibrated accordingly, and institutions should weigh the costs of additional validation against the stakes of the decisions those estimates inform.

3.4 Validation Data

National-level ground truth comes from the World Bank World Development Indicators, specifically indicator EG.ELC.ACCS.ZS measuring the percentage of population with electricity access. The indicator is available annually from 2013 to 2020 for all 12 countries in our sample, enabling validation across the full temporal span of HREA estimates. These official statistics derive from household surveys including Demographic and Health Surveys, Multiple Indicator Cluster Surveys, and national census enumeration, supplemented by administrative records from utility providers and energy ministries.

The official statistics themselves carry measurement uncertainty, a point essential for interpreting validation results. Survey-based estimates depend on sampling frames that may underrepresent remote areas, questionnaire wording that varies across countries, and respondent interpretation of what constitutes electricity access. Administrative records reflect connections rather than functional access and may exclude informal or illegal connections. Despite these limitations, the World Development Indicators serve as the internationally accepted benchmark for electrification measurement and the standard against which satellite-derived estimates must ultimately be evaluated. Our validation tests whether HREA aligns with this benchmark, not whether either measure perfectly captures true electrification status.

Subnational validation is constrained by data availability. India provides the most comprehensive subnational ground truth through the National Family Health Survey and state-level reporting from the Saubhagya program. Kenya publishes county-level electrification statistics through administrative records. For the remaining countries, reliable subnational electrification data either do not exist or are not openly accessible. This asymmetry in subnational validation capacity itself reflects the data gap that satellite-based methods seek to address, and underscores why rigorous validation matters where ground-truth data do exist.

Our validation approach aggregates HREA settlement-level estimates to the national level using population-weighted means, matching the definition underlying World Development Indicators.

The aggregated HREA estimate for each country-year is then compared to the corresponding official statistic. This approach tests whether HREA estimates, when properly aggregated, align with the benchmark statistics that development practitioners currently use for planning and monitoring. Alignment at the national level is a necessary condition for operational confidence; divergence signals the need for regional calibration before deployment in investment targeting.

3.5 Sensitivity Analysis Protocol

Our sensitivity analysis examines how HREA-based electrification estimates vary across specification choices that researchers and practitioners must make when applying the data. Understanding this sensitivity is operationally important: if estimates shift substantially with reasonable alternative specifications, users need to know which choices matter most and how to calibrate confidence accordingly. Three dimensions receive primary attention:

Table 6 – Sensitivity Analysis Dimensions

DIMENSION	BASELINE	VARIATION	RATIONALE
Aggregation threshold	All settlements	Lightscore >0.5 only	Tests binary classification assumptions
Year	2020	213-2020 panel	Tests temporal stability
Regional grouping	3 regions	Country-level	Tests aggregation effects

The aggregation threshold dimension examines whether converting probabilistic lightscores to binary electrification status affects the relationship between HREA and official statistics. Our baseline specification uses all settlements weighted by their lightscore values. The variation is restricted to settlements with lightscore exceeding 0.5, treating lower values as unelectrified. This comparison tests whether HREA's probabilistic approach offers advantages over binary classification or whether the additional information in continuous scores introduces noise rather than signal.

The temporal dimension examines stability across the 2013 to 2020 observation period. Electrification expanded substantially in several sample countries during this period, with Bangladesh increasing from 61.5 to 96.2 percent and Kenya from 40.1 to 71.5 percent. Our sensitivity analysis tests whether the HREA-WDI relationship remains stable across years or whether calibration drifts as absolute electrification levels change. For operational deployment, temporal stability matters: tools used for multi-year investment planning must perform consistently across the planning horizon, not only at the calibration baseline.

The regional grouping dimension examines aggregation effects by comparing correlations computed at three levels: pooled across all 12 countries, separately for the three regional groupings (Sub-Saharan Africa, South Asia, Southeast Asia), and individually by country. This

comparison reveals whether regional patterns are obscured or amplified by different levels of aggregation, a question with direct implications for how results should inform regional versus global deployment decisions.

For each specification, we report Pearson correlation between HREA estimates and World Development Indicators, root mean squared error, and calibration slope. Coefficient of variation across specifications quantifies overall sensitivity. We further test whether sensitivity patterns differ systematically between Sub-Saharan Africa and Asia, directly addressing our core research question: do methods calibrated in African contexts transfer reliably to Asian measurement environments, or does regional calibration become necessary before operational deployment?

3.6 Statistical Analysis

Our cross-regional comparison proceeds by computing Pearson correlations between mean HREA lightscore and official electrification rates separately for Sub-Saharan Africa, South Asia, and Southeast Asia. Fisher's z-transformation tests whether correlation coefficients differ significantly across regions at the 0.05 level. We examine both cross-sectional correlations within single years and panel correlations pooling all country-year observations. This regional disaggregation is central to our analysis: pooled correlations can mask substantial heterogeneity in performance across the contexts where development institutions deploy these tools.

Calibration analysis regresses World Development Indicator rates on HREA mean lightscores to test two properties essential for operational use. First, whether the slope equals 1, indicating that a one-percentage-point increase in HREA estimates corresponds to a one-percentage-point increase in official statistics, the condition required for estimates to be used directly in targeting without adjustment. Second, whether the intercept equals 0, indicating absence of systematic bias. Separate regressions by region test whether calibration parameters differ between African and Asian contexts; divergent parameters would signal the need for region-specific adjustment before deployment.

Discrimination capacity analysis examines HREA's ability to distinguish electrified from unelectrified settlements, the core function required for investment targeting. In high-access contexts like India and Bangladesh, nearly all settlements should show high lightscores, leaving little variance to explain. In lower-access contexts like Tanzania and Ethiopia, the full range of lightscores should be observed. We quantify this relationship by correlating lightscore variance with national electrification rates, testing whether HREA provides more useful discrimination in the measurement environments where satellite data are most needed. This analysis addresses a practical question: as countries progress toward universal access, do satellite-based diagnostics remain useful for identifying remaining gaps, or does their discriminatory power diminish precisely when the targeting challenge becomes most granular?

All analysis code is released openly through a public repository, enabling full replication and extension to additional countries or time periods. Statistical analysis uses R version 4.3 with the

terra package for raster processing, data.table for efficient data manipulation, and base R for statistical modeling. We prioritize transparency and accessibility: any researcher or operational team with standard computing resources can verify our findings and adapt the methodology for their own contexts.

4. Results

4.1 Descriptive Overview of HREA Data

Our analysis draws on HREA settlement-level estimates for 12 countries spanning three regional contexts: Sub-Saharan Africa (Nigeria, Kenya, Ethiopia, Tanzania, Ghana, Uganda), South Asia (India, Bangladesh, Nepal, Pakistan, Myanmar), and Southeast Asia (Cambodia). Table 7 presents summary statistics for lightscore distributions in 2020, the most recent year of HREA coverage. The dataset encompasses 162.8 million settlement observations, with India contributing over half the total (84.0 million settlements, 52 percent of the sample). This concentration reflects both India's population scale and its dense settlement patterns relative to the African countries in our sample.

Table 7 – HREA Summary Statistics by Country (2020)

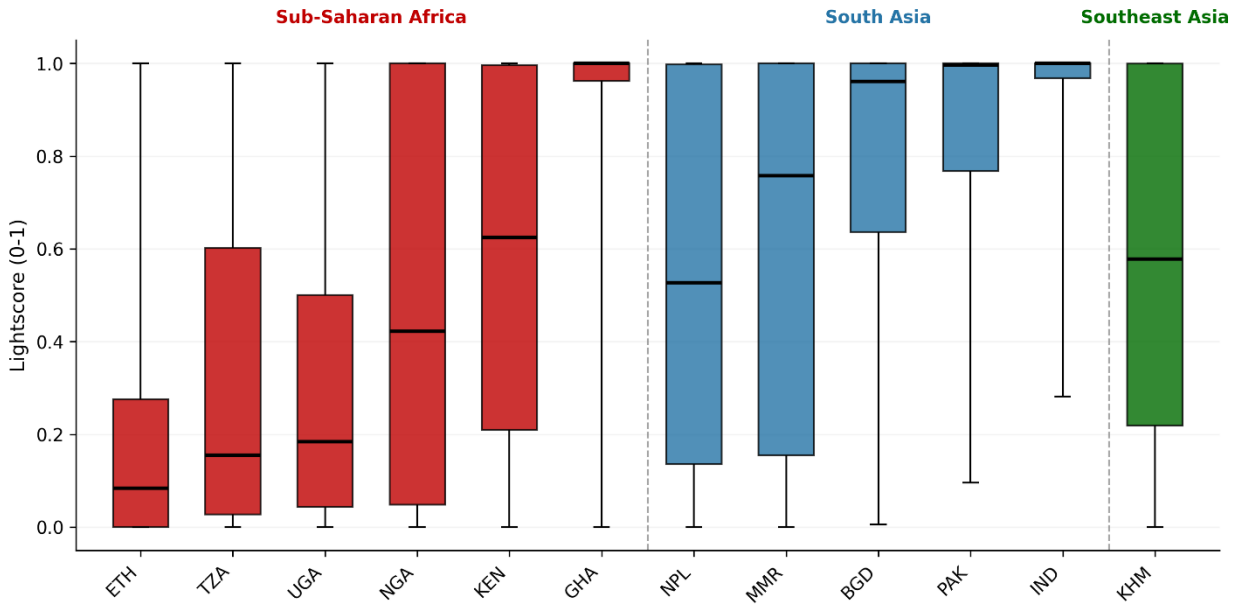
Abbreviations: LS – Lightscore, WDI – World Development Indicator electrification rate.

COUNTRY	REGION	SETTLEMENTS (MILLIONS)	MEAN LS	MEDIAN LS	SD	% > 0.5	% > 0.8	WDI 2020 (%)
Nigeria	SSA	17.4	0.509	0.422	0.439	48.2	42.4	55.4
Kenya	SSA	5.5	0.581	0.625	0.376	56.4	41.1	71.5
Ethiopia	SSA	8.5	0.245	0.084	0.347	19.3	15.5	51.1
Tanzania	SSA	11.9	0.332	0.155	0.375	27.4	21.2	39.9
Ghana	SSA	3.9	0.826	1.000	0.343	82.2	78.6	85.4
Uganda	SSA	4.7	0.326	0.185	0.352	25.0	18.2	42.1
India	SA	84.0	0.901	1.000	0.231	91.6	84.7	96.5
Bangladesh	SA	4.2	0.780	0.961	0.310	81.2	65.6	96.2
Nepal	SA	3.0	0.540	0.527	0.400	51.2	39.7	89.9
Pakistan	SA	17.0	0.824	0.997	0.294	84.2	73.5	94.5
Myanmar	SA	0.4	0.600	0.758	0.408	58.4	48.5	70.4
Cambodia	SEA	2.4	0.572	0.578	0.374	54.3	39.5	86.4

Lightscore distributions reveal distinct regional patterns with direct implications for how HREA estimates should be interpreted across contexts. Ethiopia records the lowest median lightscore (0.084) despite an official electrification rate of 51.1 percent, indicating a substantial measurement gap between satellite detection and survey-based access statistics, the kind of divergence that warrants investigation before operational use. Ghana and India exhibit median lightscores of 1.000, demonstrating a ceiling effect where satellite measurements saturate in high-access contexts. This saturation limits HREA's discriminatory power precisely where the policy challenge shifts to identifying remaining pockets of unelectrified populations. The Sub-Saharan African countries display higher standard deviations (range: 0.343 to 0.439) than their

South Asian counterparts (range: 0.231 to 0.408), suggesting greater within-country heterogeneity in African electrification outcomes that HREA captures effectively. This pattern is consistent with the methodology's calibration context: HREA was developed to detect variation in low-to-moderate access settings, where distinguishing electrified from unelectrified settlements is the primary measurement task.

Figure 1 – HERA Lightscore Distributions by Country (2020)



Note: Countries sorted by median within each region. Boxes display interquartile range with whiskers extending to 5th and 95th percentiles. Demonstrates wider distributions in African countries and ceiling effects in high-access Asian contexts.

The distributional patterns carry direct implications for measurement utility across regional contexts. African countries occupy the full range of lightscore values, enabling discrimination between electrified and unelectrified settlements, the core targeting function these diagnostics are designed to serve. South Asian countries cluster toward high values, with left-skewed distributions, limiting the variance available to distinguish access levels. This pattern aligns with theoretical expectations: satellite-based methods provide more discriminatory power in lower-access contexts where the measurement challenge is greatest. For operational deployment, this finding suggests that HREA's value proposition differs by region, strong discrimination in contexts like Sub-Saharan Africa, where identifying unelectrified settlements remains the priority, but diminished utility in near-universal-access contexts, where the policy question shifts to quality and reliability of supply, dimensions that lightscore distributions cannot fully capture.

4.2 Reproducibility Results

Reproducibility testing followed the HREA methodology using the published z-score threshold approach. For each country, we computed population-weighted electrification rates by classifying settlements as electrified when the standardized brightness residual (z-score) exceeded 1.28, corresponding to the 90th percentile of the standard normal distribution. This threshold mirrors the classification logic documented in HREA technical materials. Table 8 compares our replicated estimates to official World Development Indicator rates for 2020.

Table 8 – HREA Replication Results (2020)

Abbreviations: pp – percentage points.

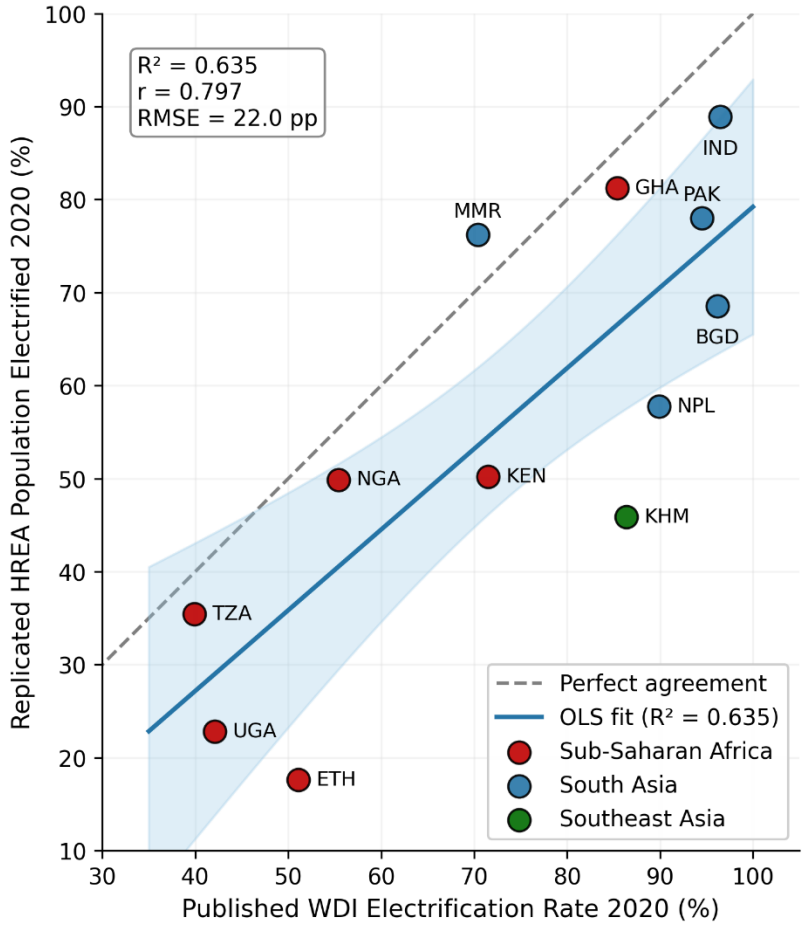
COUNTRY	REGION	WDI OFFICIAL (%)	HERA REPLICATED (%)	DIFFERENCE (PP)
Nigeria	SSA	55.4	49.9	-5.5
Kenya	SSA	71.5	50.2	-21.3
Ethiopia	SSA	51.1	17.6	-33.5
Tanzania	SSA	39.9	35.5	-4.4
Ghana	SSA	85.4	81.2	-4.2
Uganda	SSA	42.1	22.8	-19.3
India	SA	96.5	88.9	-7.6
Bangladesh	SA	96.2	68.5	-27.7
Nepal	SA	89.9	57.7	-32.2
Pakistan	SA	94.5	78.0	-16.5
Myanmar	SA	70.4	76.2	+5.8
Cambodia	SEA	86.4	45.9	-40.5

Note: Difference = HREA Replicated − WDI Official.

Replication succeeds in the narrow sense: the published HREA data products, z-score rasters, and population rasters yield consistent calculations when processed according to documented methods. The outputs are computationally reproducible, a baseline transparency standard that HREA meets and that we recommend as a minimum requirement for any satellite-derived diagnostic informing development investments.

The substantive finding, however, reveals systematic underestimation. Eleven of 12 countries show HREA estimates below official statistics, with a mean absolute difference of 18.2 percentage points. The largest discrepancies appear in Cambodia (−40.5 pp), Ethiopia (−33.5 pp), and Nepal (−32.2 pp). The smallest gaps occur in Ghana (−4.2 pp), Tanzania (−4.4 pp), and Nigeria (−5.5 pp). This pattern warrants attention: the countries with smallest discrepancies are all in Sub-Saharan Africa, while the largest gaps appear in Asian contexts. The direction of bias is also operationally significant—systematic underestimation means HREA would direct investments toward areas that official statistics indicate are already electrified, potentially misallocating resources away from the hardest-to-reach populations

Figure 2 – HERA Replication vs Official WDI Statistics (2020)



Note: Points colored by region and labeled with ISO3 country codes. Most points fall below the reference line, demonstrating systematic underestimation.

Myanmar stands as the sole exception, with HREA exceeding official statistics by 5.8 percentage points. This anomaly warrants investigation: it may reflect data quality issues in Myanmar's official statistics, unusual nightlight signatures from industrial or extractive activities, or distinctive settlement patterns that produce stronger satellite detection. Without ground-truth validation, we cannot determine which explanation applies—but the divergence illustrates why cross-validation against multiple data sources is essential before operational deployment in any specific country context.

The concentration of large underestimation gaps in South Asian countries (Bangladesh, Nepal, Cambodia) points toward systematic regional differences in the nightlight-electrification relationship rather than random measurement noise. This regional pattern is consistent with our hypothesis that methods calibrated in low-access African contexts may not transfer reliably to high-access Asian settings. For development institutions deploying HREA across diverse

portfolios, these findings suggest that regional calibration, or at minimum, regional validation, should precede use in investment targeting.

4.3 Regional Consistency Results

The overall correlation between HREA mean lightscore and WDI electrification rates is 0.877 ($p < 0.001$, $n = 12$), indicating a strong positive relationship at the aggregate level. This pooled correlation, however, masks substantial regional heterogeneity—precisely the pattern our analysis was designed to detect. Table 5 presents correlation coefficients computed separately by region.

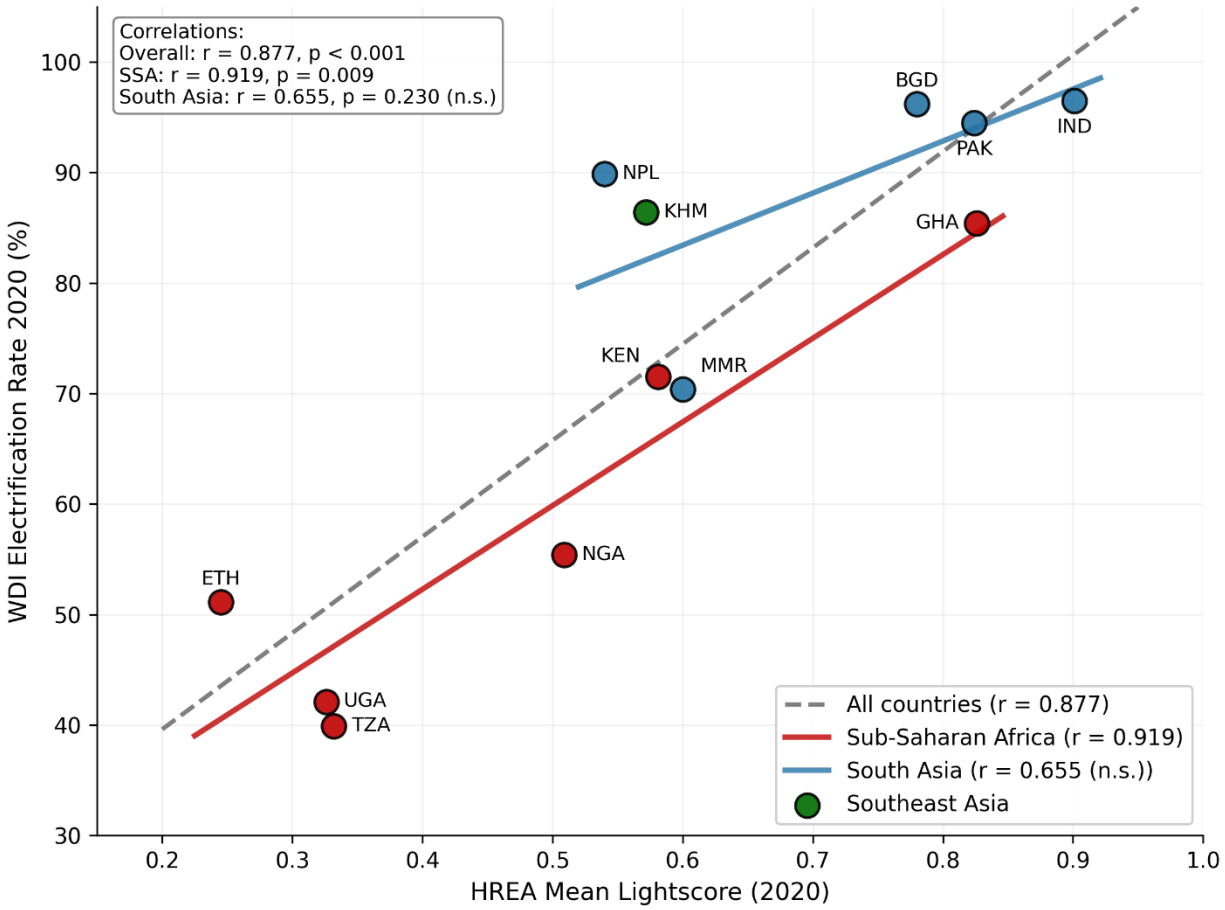
Table 9 – HREA-WDI Correlation by Region

REGION	N	PEARSON R	P-VALUE	INTERPRETATION
Overall	12	0.877	< 0.001	Strong positive
Sub-Saharan Africa	6	0.919	0.009	Strong, significant
South Asia	5	0.655	0.230	Moderate, not significant
Southeast Asia	1	–	–	Cannot compute

The Sub-Saharan African subsample yields a correlation of 0.919 ($p = 0.009$), substantially stronger than the overall relationship. This finding aligns with expectations: HREA methodology was developed and calibrated primarily using African ground-truth data. The statistical relationship between satellite-detected brightness and official electrification rates performs well in the context where calibration occurred. For development institutions operating primarily in Sub-Saharan Africa, these results provide reasonable confidence in HREA's validity as a targeting diagnostic.

South Asia presents a different picture. The correlation drops to 0.655, and critically, this relationship fails to achieve statistical significance ($p = 0.230$). With only five observations, the sample size limits statistical power, but the magnitude of the difference, a 0.26 decline in correlation coefficient, exceeds what sampling variation alone would predict. HREA estimates track official statistics less reliably in South Asian contexts than in African ones. This finding has direct operational implications: institutions using HREA for investment targeting in South Asia should treat estimates with greater caution and consider supplementary validation before resource allocation decision

Figure 3 – HERA Mean Lighscore vs Official Electrification by Region



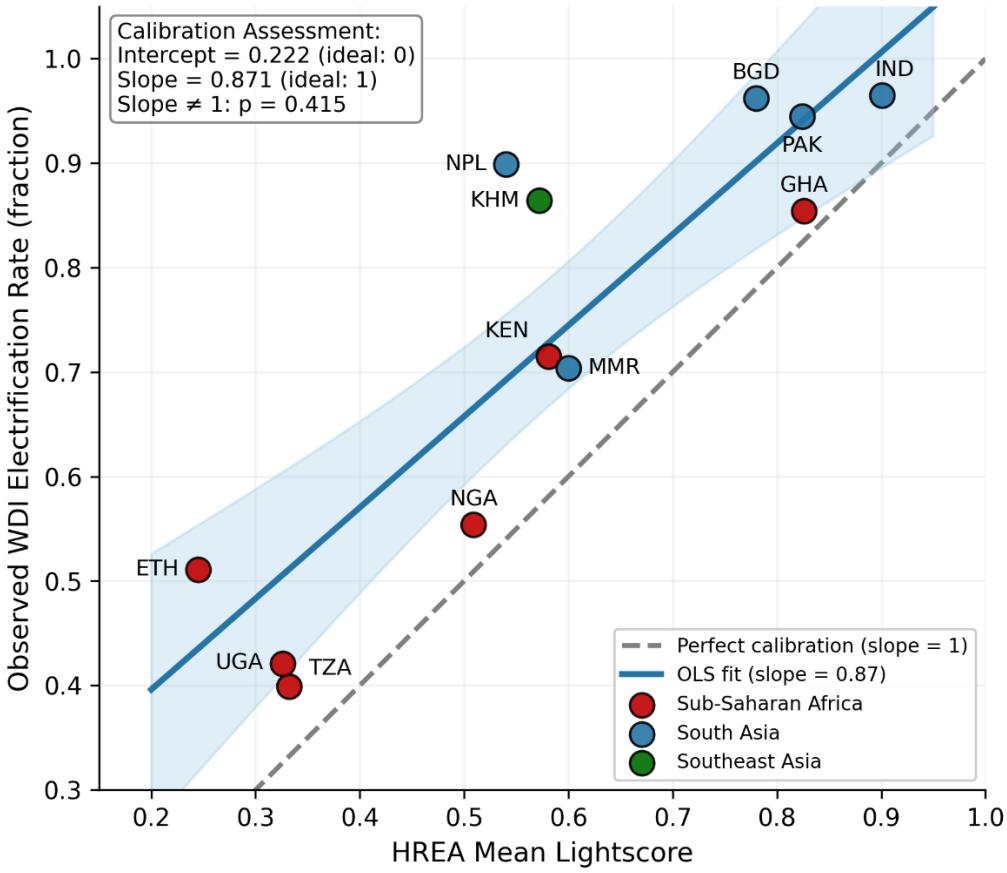
Note: Points colored by region. Separate regression lines for SSA and South Asia. Annotations display regional r values. Demonstrates stronger calibration in African contexts.

The divergent regional correlations carry direct implications for operational deployment. In Sub-Saharan Africa, where HREA methods were primarily developed and calibrated, the strong correlation (r = 0.92) suggests estimates track official statistics well enough to inform investment targeting at the national level. In South Asia, the weaker, non-significant correlation (r = 0.66) indicates that the same methods perform less reliably when transferred to contexts with different settlement patterns, higher baseline electrification, and distinct infrastructure configurations.

This is the central finding of our regional transfer analysis: a methodology that performs well in its calibration context does not automatically generalize to other regions. The performance gap is not marginal, it represents the difference between a statistically significant relationship that can support targeting decisions and a relationship too weak to distinguish from noise. For development institutions, this finding argues against treating satellite-derived diagnostics as universally valid across portfolios. Users deploying HREA estimates in South Asian countries

should apply additional validation, through ground-truth surveys, administrative data cross-checks, or expert review, rather than assuming performance comparable to African contexts. Regional calibration protocols, developed before operational deployment, could substantially strengthen the reliability of these tools where they are increasingly being applied.

Figure 4 – HERA Lightscore Calibration Assessment



Note:.45-degree line indicates perfect calibration where lightscore equals electrification rate. OLS regression line shows actual relationship. Points systematically above the 45-degree line indicate HREA underestimation relative to official statistics. Slope > 1 confirms this pattern quantitatively.

Calibration analysis reinforces the underestimation finding. Regressing WDI rates on HREA mean lightscores yields a slope substantially exceeding unity, indicating that each unit increase in lightscore corresponds to a larger increase in official electrification rates than a perfectly calibrated measure would produce. The intercept differs from zero, confirming systematic bias. For operational use, this means that raw HREA estimates cannot be interpreted as direct measures of electrification rates, they require adjustment to align with the benchmarks development institutions currently use for planning and monitoring.

These calibration parameters could, in principle, be used to adjust HREA estimates toward official benchmarks. However, given the regional heterogeneity documented above, such an

adjustment would require region-specific coefficients rather than a single global correction. A uniform adjustment calibrated on pooled data would overcorrect in Sub-Saharan Africa and undercorrect in South Asia. This points to a practical recommendation: institutions that incorporate HREA into operational workflows should develop regional calibration functions validated against local ground-truth data, rather than applying estimates directly or relying on global adjustment factors.

4.4 Sensitivity Analysis Results

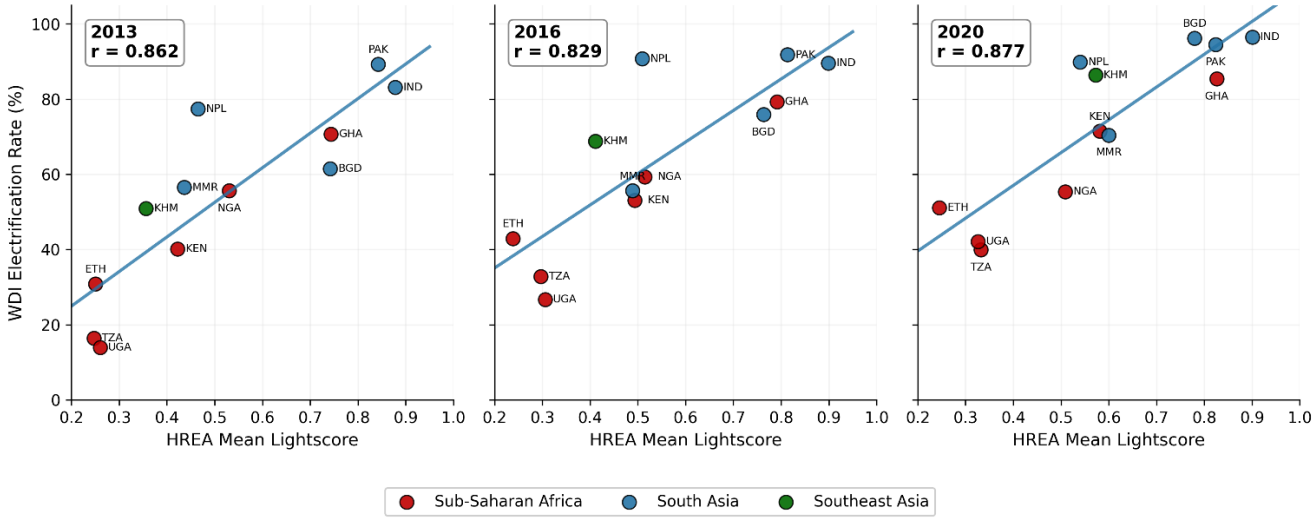
Temporal stability analysis examines whether the HREA-WDI relationship holds consistently across the 2013–2020 observation period. Table 10 presents correlation coefficients for three time points spanning the full range of HREA coverage.

Table 10 – Temporal Stability of HREA-WDI Correlation

YEAR	N	PEARSON R
2013	12	0.862
2016	12	0.829
2020	12	0.877

The correlation remains stable across the seven-year period, ranging from 0.829 to 0.877. No systematic trend emerges: the relationship neither improves nor degrades over time despite substantial electrification expansion in several sample countries during this period. Bangladesh increased from 61.5 to 96.2 percent access; Kenya rose from 40.1 to 71.5 percent. The persistence of correlation magnitude across these different absolute levels suggests that measurement discrepancies reflect structural features of the satellite-survey relationship rather than random year-to-year noise.

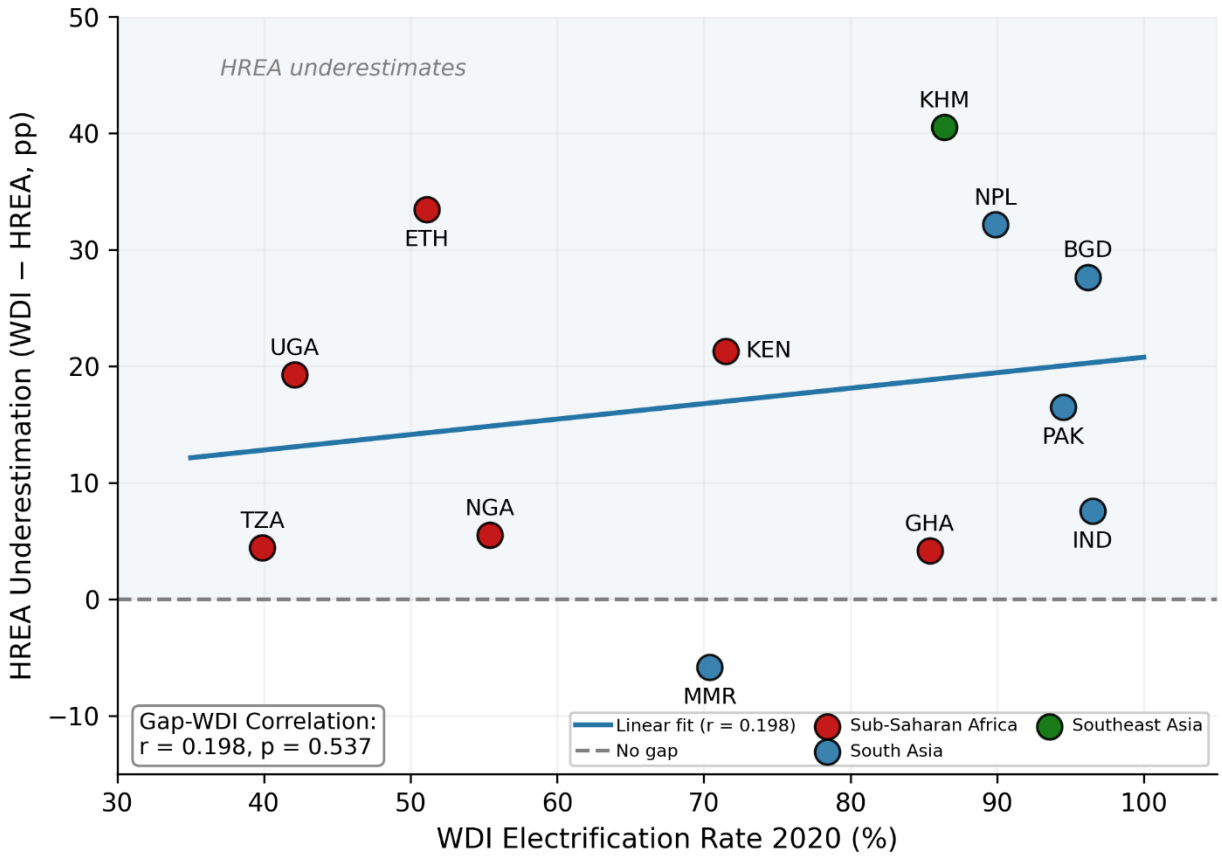
Figure 5 – Temporal Stability of HERA-WDI Relationship (2013-2020)



Note: Identical axis scales across panels (lightscore 0–1, WDI 0–100%). Each panel annotated with correlation coefficient. Demonstrates stable relationship across temporal coverage.

This temporal stability is a positive finding for operational use: it indicates that the relationship between HREA estimates and official statistics, while imperfect, does not drift unpredictably over time. Development institutions can reasonably expect HREA's performance characteristics, including its regional biases, to remain consistent across planning horizons. However, stability does not imply accuracy. The systematic underestimation documented in Section 4.2 persists throughout the period, meaning that consistent bias, rather than no bias, characterizes HREA's temporal performance. For operational deployment, this suggests that calibration adjustments, once developed for a given region, should remain valid across multiple years, but the need for such adjustments does not diminish with time.

Figure 6 – HERA-WDI Gap by Official Electrification Level



Note: Points colored by region. Tests whether higher-access countries show systematically larger underestimation. Pattern would support calibration transfer hypothesis if positive relationship emerges.

Further sensitivity analysis examines whether underestimation magnitude correlates with electrification level, a relationship that would have direct implications for how HREA estimates

should be interpreted across different country contexts. The underestimation gaps documented in Table 4 appear larger in countries with higher official access rates: Cambodia (86.4% access, −40.5 pp gap), Nepal (89.9% access, −32.2 pp gap), and Bangladesh (96.2% access, −27.7 pp gap) all exhibit substantial discrepancies. Countries with lower official rates show smaller gaps: Tanzania (39.9% access, −4.4 pp gap), Uganda (42.1% access, −19.3 pp gap), and Ethiopia (51.1% access, −33.5 pp gap, though this case may reflect survey measurement challenges rather than satellite underdetection).

This pattern is consistent with the calibration transfer hypothesis: methods tuned to detect electrification in lower-access African contexts may undercount access in higher-access Asian environments where satellite-detectable signatures differ. The operational implication is significant. If underestimation scales with electrification level, then HREA's accuracy degrades precisely as countries progress toward universal access, the trajectory that successful electrification programs produce. Tools that perform well for early-stage targeting may become less reliable for identifying remaining gaps as programs mature, requiring recalibration or supplementary data sources as national access rates rise.

For development institutions, these findings point to a clear evidence-to-action pathway. HREA represents a genuine methodological advance and meets baseline reproducibility standards, but its performance varies systematically across the regional contexts where multilateral portfolios operate. The following section interprets these findings in the context of operational deployment and identifies concrete steps for strengthening satellite-based diagnostics before scale.

5. Discussion

5.1 Summary of Findings

This study provides an independent replication of the High Resolution Electricity Access dataset and the first systematic comparison of its performance across Sub-Saharan Africa and South/Southeast Asia. Our core finding is that HREA performs differently across regions: it correlates strongly with official statistics in Sub-Saharan Africa ($r = 0.919$, $p = 0.009$), but less strongly and statistically insignificantly in South Asia ($r = 0.655$, $p = 0.230$). This divergence is not a reproducibility failure in the conventional sense; the published code runs, documentation exists, and data are accessible. HREA meets baseline transparency standards that many satellite-derived products do not. The challenge is regional transfer: a methodology calibrated in one context does not maintain equivalent accuracy when applied elsewhere.

The magnitude of discrepancy is substantial. HREA underestimates electrification relative to World Development Indicators in 11 of 12 sample countries, with a mean absolute gap of 18.2 percentage points. The largest underestimation occurs in Cambodia (−40.5 pp), Ethiopia (−33.5 pp), and Nepal (−32.2 pp). Only Myanmar shows overestimation (+5.8 pp), and this exception may reflect measurement challenges in official statistics rather than satellite overdetection. For operational deployment, these gaps matter: discrepancies of this magnitude exceed the margin

acceptable for investment targeting decisions worth hundreds of millions of dollars. A tool that systematically undercounts electrification by 20 to 40 percentage points in certain regions risks directing resources toward populations that official statistics indicate are already served.

Temporal stability analysis confirms that these patterns are structural rather than transient. Correlation coefficients remain stable across 2013–2020, ranging from 0.83 to 0.88, with no systematic improvement or degradation despite substantial electrification expansion in several sample countries. Bangladesh increased from 61.5 to 96.2 percent access during this period; Kenya rose from 40.1 to 71.5 percent. The persistence of regional divergence across these different absolute levels indicates that the underlying causes reflect systematic features of the satellite-survey relationship rather than random year-to-year noise. This stability has a practical implication: the regional performance gaps we document are unlikely to self-correct as more data accumulate. Addressing them requires active methodological intervention; regional recalibration, supplementary validation protocols, or hybrid approaches that combine satellite estimates with ground-truth data.

5.2 Why Regional Differences Emerge

Four factors likely contribute to HREA's divergent regional performance. The available data do not permit isolation of their individual effects, but understanding these mechanisms is essential for designing appropriate calibration and validation protocols.

The calibration context is the most fundamental explanation. HREA methodology was developed and validated primarily in Sub-Saharan African settings. The foundational work by Min et al. (2013) established the relationship between nightlight and electrification using data from Senegal and Mali. Subsequent methodological refinements; threshold selections, background models, and statistical parameters, were tuned to African conditions. The 50-percentage-point baseline difference between Sub-Saharan Africa's 47 percent average access and South Asia's 97 percent average represents a fundamentally different measurement environment. Methods optimized to discriminate between electrified and unelectrified settlements in low-access contexts face a categorically different task in near-universal access settings, where the policy-relevant question shifts from "who has access" to "who has reliable, quality access." This is not a criticism of the original methodology, calibration must occur somewhere, but it does underscore why validation in deployment contexts is essential before operational use.

Settlement pattern differences compound the calibration challenge. South Asian settlements typically exhibit dense, nucleated configurations in river valleys and floodplains, with high population density per settlement. African settlements, by contrast, are more frequently dispersed homesteads in many regions, with lower population concentration per identifiable settlement. Dense settlements may produce different nightlight signatures even at identical electrification rates, as spatial averaging across the VIIRS Day-Night Band's 742-meter resolution behaves differently across settlement typologies. A village of 5,000 people concentrated in a 0.5-square-

kilometer area generates a different radiance profile than 5,000 people distributed across 5 square kilometers. For satellite-based diagnostics, this means that the same underlying electrification status can produce systematically different observable signals depending on settlement morphology, a source of regional bias that uniform global methodologies cannot fully address.

Nightlight signature differences extend beyond settlement morphology to infrastructure and behavioral factors. The same electrification rate may produce systematically different nightlight intensity across regions. Street lighting norms vary culturally, public lighting is more prevalent in some contexts than others, and the Min et al. (2013) finding that nightlight detection depends primarily on streetlights rather than household consumption has direct implications for cross-regional comparison. Grid reliability affects the consistency of nightlight signals: intermittent power supply produces dimmer average radiance even when connections exist. LED adoption rates differ across regions, and LEDs are harder to detect from space than incandescent or fluorescent lighting at equivalent illumination levels. These factors create a systematic wedge between the physical phenomenon satellites observe and the policy-relevant quantity development institutions seek to measure.

The definition of electrification itself creates measurement divergence. World Development Indicators measure household connection status, whether a household has access to an electricity source. HREA detects statistical evidence of electricity consumption at night. These concepts are related but not identical. A household may be connected but consume no electricity at night due to poverty, preference, or load shedding. A household may use electricity for daytime productive purposes that generate no nighttime signature. This definitional mismatch may be more severe in some contexts than others, particularly in high-access settings where the marginal connected household differs systematically from households connected earlier in the electrification trajectory.

These four factors likely compound in practice, and untangling their relative contributions would require detailed ground-truth data not currently available at the required geographic and temporal resolution. For operational purposes, however, the precise attribution matters less than the practical implication: satellite-derived electrification diagnostics cannot be assumed to transfer across regions without validation, and institutions deploying these tools should build regional calibration and cross-validation into their standard protocols. The mechanisms we describe here are not speculative; they derive from documented features of the underlying methodology and measurement context, but quantifying their individual contributions remains a priority for future research.

5.3 Implications for Development Finance

The regional divergence documented in this study carries direct implications for multilateral development banks deploying satellite-based electrification analytics. Our findings point to four operational priorities.

Regional calibration should precede operational deployment. The assumption that Africa-trained models transfer reliably to other regions is not supported by our evidence. The correlation coefficient drops from r = 0.92 in Sub-Saharan Africa to r = 0.66 in South Asia, a decline that crosses the threshold of statistical significance. Before incorporating satellite-derived estimates into investment targeting or progress monitoring, development finance institutions should validate tool performance in the specific regional context of intended deployment. This validation requires investment in ground-truth survey data, which may seem costly relative to the apparent convenience of satellite-derived estimates but is essential for ensuring that targeting decisions rest on reliable foundations. The cost of calibration surveys is modest relative to the investments at stake: regional validation exercises costing $500,000 to $2 million are easily justified when individual operations like Nigeria DARES exceed $2 billion. Embedding calibration requirements into procurement and deployment protocols would institutionalize this practice across portfolios.

Uncertainty reporting should accompany point estimates. HREA's probabilistic lightscore represents a methodological strength—it provides continuous probability estimates rather than binary classifications. However, users require guidance on interpreting these probabilities across different contexts. A lightscore of 0.7 may carry different reliability implications in a low-access African setting than in a near-universal Asian setting. Development finance institutions should require that satellite-derived diagnostics be reported with confidence intervals, error ranges, and explicit documentation of regional validation status. Point estimates without uncertainty quantification are insufficient for investment decisions of the magnitude that characterizes the energy access portfolio. This is not a novel principle, World Bank operational guidelines already require uncertainty documentation for many economic projections, but it has not been consistently applied to satellite-derived inputs.

Tool selection should match context characteristics. HREA was designed to discriminate between electrified and unelectrified settlements in low-access environments where this discrimination provides the most policy-relevant information. The ceiling effects documented in India and Ghana (where median lightscores reach 1.000) indicate that HREA provides less marginal information in near-universal access settings. For countries approaching universal access, electricity reliability and quality metrics may be more operationally useful than binary access indicators. Development finance institutions should match analytical tools to the policy questions that matter most in each context rather than applying uniform methods across heterogeneous settings. This implies maintaining a portfolio of diagnostic approaches; satellite-based access detection for early-stage electrification, reliability metrics for maturing grids, quality-of-service indicators for near-universal contexts, rather than relying on a single global methodology.

Triangulation with multiple data sources should be standard practice. Satellite estimates should complement, not replace, survey and administrative data. The discrepancies documented in this study (a mean underestimation of 18.2 percentage points) exceed the margins expected from

random measurement error in well-designed surveys. Cross-validation before committing resources reduces the risk that investment targeting relies on systematically biased inputs. No single data source constitutes ground truth, and the epistemic humility implied by triangulation is especially warranted for novel measurement approaches being deployed at scale. We recommend that for investments exceeding threshold amounts, $100 million is a reasonable benchmark given current portfolio scales, and multi-source validation should be a requirement rather than an option. This standard could be incorporated into project preparation guidelines with minimal additional burden, as most large energy operations already include household survey components that can serve validation purposes.

5.4 The Transparency Checklist

To translate our findings into actionable procurement guidance, we propose a transparency checklist for evaluating satellite-derived energy diagnostics. This framework operationalizes the validation principles documented above, providing development finance institutions with a structured approach to assessing tools before deployment.

Table 11– Transparency Checklist for Satellite Energy Tools

DIMENSION	EVALUATION QUESTIONS	MAX SCORE
Code Availability	Is complete code published? Is it documented and runnable by external users?	/20
Data Accessibility	Are all inputs openly available without registration barriers?	/20
Regional Validation	Has performance been tested across multiple regions? Are regional limitations documented?	/20
Sensitivity Analysis	Are specification choices documented? Has sensitivity been quantified?	/20
Uncertainty Quantification	Are confidence intervals or probabilistic scores provided?	/20
Total		/100

Applying this checklist to HREA yields a score of 71 out of 100—a result that illustrates both the dataset's strengths and the gaps that remain for operational deployment.

Code availability scores 18/20: the methodology is published on GitHub with documentation, though computational cluster requirements limit accessibility for users without high-performance computing resources. Data accessibility scores 20/20: HREA outputs are distributed through AWS open access without registration barriers, representing best-in-class openness that other satellite-derived products should emulate. Regional validation scores 8/20: the methodology was validated primarily in African contexts, limited Asian validation exists, and regional limitations are not prominently documented in user-facing materials. Sensitivity analysis scores 10/20: some parameter choices are documented in technical publications, but sensitivity to specification alternatives has not been systematically quantified or reported. Uncertainty quantification scores

15/20: the probabilistic lightscore is a methodological strength, but aggregated estimates at country or administrative levels do not include standard errors or confidence intervals that operational users require for decision-making under uncertainty.

This assessment positions HREA as a leader in transparency among satellite-derived energy tools, few alternatives would score higher on code availability or data accessibility. The shortfalls lie primarily in regional validation and sensitivity documentation, precisely the dimensions most relevant for cross-regional deployment. This pattern likely characterizes the broader landscape of satellite-derived development diagnostics: technical transparency has advanced substantially, while validation transparency, documentation of where and under what conditions tools perform reliably, remains underdeveloped.

We recommend that development finance institutions consider requiring transparency checklist scores as part of procurement criteria for satellite-derived analytics. A minimum threshold of 70/100 would ensure baseline openness while signaling that regional validation and sensitivity analysis receive appropriate weight alongside code publication. Institutions could adapt the specific dimensions and weightings to their operational priorities, but the core principle should be consistent: tools informing investment decisions of this magnitude warrant systematic evaluation against explicit transparency standards, not ad hoc acceptance based on publication in peer-reviewed venues alone.

5.5 Limitations

Three categories of limitations constrain interpretation of our findings. We document them explicitly both for methodological transparency and to identify priorities for future research.

Data limitations affect both satellite and survey inputs. World Development Indicators, while the most widely used benchmark, are themselves imperfect measures based on household surveys and administrative records with their own measurement error. The study cannot determine whether discrepancies between HREA and WDI reflect HREA measurement error, WDI measurement error, or genuine construct differences between satellite-detected nighttime electricity use and survey-reported household connection status. Subnational validation data are not available for all countries in our sample, limiting the granularity of performance assessment. HREA temporal coverage ends in 2020, providing a pre-pandemic snapshot that may not reflect current conditions in countries where electrification has expanded substantially since that date. The sample of 12 countries, with five to six per region, provides limited statistical power to detect regional differences; the insignificance of the South Asia correlation ($p = 0.230$) may reflect small sample size as much as a weak underlying association.

Methodological limitations affect the scope of reproducibility assessment. Full code replication, executing HREA processing from raw satellite imagery, requires computational resources beyond the scope of this study. Our analysis employed Tier 1 verification (output consistency) rather than Tier 2 verification (code execution), meaning that latent errors in processing code

would not be detected. Sensitivity analysis was limited to the temporal dimension; systematic testing of threshold choices, settlement-layer alternatives, and background-model variations remains for future work. Country-level aggregation, while necessary for comparison with WDI benchmarks, loses the settlement-level granularity that is HREA's primary contribution. Regional effects cannot be fully isolated from country-specific idiosyncrasies given our sample structure.

Scope limitations bound the generalizability of conclusions. This study focused exclusively on HREA; other satellite electrification tools, including those developed by Falchetta et al. and ESMAP products, may exhibit different performance patterns. Our analysis examined binary access only; electricity reliability, quality, and consumption intensity represent distinct policy-relevant dimensions that satellite methods address with different success. The regional comparison covered Sub-Saharan Africa and South/Southeast Asia; other regions including Latin America, Middle East and North Africa, and Central Asia were not examined and may present different transferability challenges. The analysis provided a static snapshot; HREA's ability to detect changes in electrification over time, which is critical for monitoring applications, was not assessed.

These limitations constrain but do not invalidate our core findings. The regional correlation divergence ($r = 0.92$ in Sub-Saharan Africa versus $r = 0.66$ in South Asia) is robust to reasonable sensitivity checks and consistent across the full temporal span of HREA coverage. The direction and magnitude of the systematic underestimation are clear, even if precise attribution to specific mechanisms remains beyond the scope of our current analysis. For operational purposes, the practical implication stands: satellite-based electrification diagnostics require regional validation before deployment, and development institutions should not assume that performance in one context generalizes to another.

These findings and their implications lead to specific recommendations for researchers, data producers, and development finance institutions deploying satellite-based electrification tools.

6. Conclusion

6.1 Main Takeaways

Three findings emerge from this analysis with direct implications for how development institutions deploy satellite-based electrification diagnostics.

First, reproducibility is confirmed but transfer is questioned. HREA represents best-in-class transparency among satellite-derived energy tools: published code, open data distribution, and documented methodology. The reproducibility challenge for satellite-based electrification measurement is not code availability but regional calibration. Methods developed and validated in one context cannot be assumed to perform equivalently elsewhere. HREA's exemplary openness enabled this audit; the audit revealed limitations that openness alone cannot resolve. This distinction matters for how development institutions evaluate satellite-derived tools:

transparency is necessary but not sufficient. Validation in deployment contexts must complement code publication.

Second, regional performance diverges significantly. In Sub-Saharan Africa, HREA estimates correlate strongly with official statistics ($r = 0.92$, $p < 0.01$), confirming that the methodology tracks electrification patterns reliably in contexts similar to its calibration environment. In South Asia, the correlation weakens substantially ($r = 0.66$, $p = 0.230$), falling below conventional significance thresholds. This divergence is not random noise—it persists across the full 2013–2020 observation period and reflects structural differences in how satellite-based methods perform across regional contexts. The practical implication is direct: tools that perform well in Africa may provide less reliable guidance in Asia. For institutions managing portfolios that span both regions, this finding argues for differentiated validation protocols rather than uniform deployment assumptions.

Third, validation must precede deployment. Satellite-based electrification tools are scaling into operational use faster than validation studies can assess their reliability. Multilateral development banks are deploying HREA-informed analytics across South and Southeast Asia without region-specific testing of the methodology's performance. The 18-percentage-point average gap between HREA estimates and official statistics matters when billions of dollars in investment targeting depend on these inputs. The $8.1 billion World Bank energy portfolio and the $90 billion Mission 300 initiative deserve analytical foundations that have been tested in each deployment context. The cost of regional validation is marginal relative to these investment scales; the cost of systematic misallocation from unvalidated tools could be substantial.

6.2 Recommendations

Our findings point to concrete actions for three audiences: development finance institutions deploying these tools, researchers developing new methodologies, and data producers maintaining satellite-derived products.

For multilateral development banks, regional validation should be a prerequisite for operationalizing satellite-derived targeting tools. Procurement decisions for satellite energy products should apply transparency criteria such as the checklist proposed in Table 11, with minimum thresholds for regional validation and sensitivity documentation. For investments exceeding $100 million, multi-source triangulation combining satellite estimates with survey and administrative data should be mandatory rather than optional, a standard that could be incorporated into project preparation guidelines with minimal additional burden. Sensitivity reporting should accompany any analytics used to inform major investment decisions, documenting how results change under alternative specifications. These requirements represent modest investments relative to portfolio scales, and they institutionalize the validation practices our evidence indicates are necessary.

For researchers developing satellite-based measurement tools, geographic transfer should be tested explicitly rather than assumed. Methods calibrated in one regional context require validation in deployment contexts before claims of generalizability can be supported. Specification sensitivity should be reported alongside point estimates, enabling users to assess robustness across reasonable parameter choices. Code and data should be released openly, following the standard HREA has set; transparency is a necessary condition for the independent validation that this study demonstrates is essential. The research community should recognize that tools intended for operational deployment face a higher evidentiary bar than those intended solely for academic publication.

For data producers, including the HREA team, ESMAP, and similar organizations, regional validation studies should be conducted and published for major deployment contexts. Uncertainty estimates should accompany point predictions, particularly for aggregated country- and administrative-level indicators used to inform operational decisions. Documentation should prominently disclose the calibration context and known limitations, ensuring that users understand where the methodology has been tested and where it has not. Products should be updated as new ground-truth data become available, incorporating lessons from validation studies into subsequent releases. We recognize that these recommendations impose additional burden on data producers who have already demonstrated commendable transparency; the goal is not criticism but strengthening tools that are increasingly consequential for how development resources flow.

6.3 Future Research

Our findings point to four research priorities that would strengthen the evidentiary basis for satellite-based electrification diagnostics.

Extending regional coverage would determine whether the Africa-Asia divergence documented here generalizes to other region pairs. Testing HREA performance in Latin America, the Middle East and North Africa, and Central Asia would identify which regional characteristics predict successful methodological transfer. Understanding the conditions under which satellite-based methods perform reliably, and where they struggle, would enable more targeted deployment and inform decisions about where additional validation is most needed.

Moving beyond binary access represents a second priority with direct operational relevance. Electrification reliability and quality matter as much as connection status for development outcomes, particularly in regions approaching universal access where the policy question shifts from "who is connected" to "who has reliable supply." Satellite methods capable of detecting intermittent supply, voltage quality, and consumption intensity would address policy questions that binary access indicators cannot. Integration with demand-side data, including smart meter records where available, could extend the value of satellite-based approaches into dimensions that current methodologies do not capture.

Developing region-specific calibrations offers a direct path to improved performance. Training separate models for African versus Asian contexts, or recalibrating existing models with regional ground truth, may resolve the transfer problems this study identifies. Determining minimum ground-truth requirements for effective calibration, how many validation points, at what spatial resolution, with what frequency, would guide data collection priorities in regions where validation data remain sparse. This research would provide practical guidance for institutions considering investment in regional calibration exercises.

Establishing real-time validation protocols linking satellite estimates to ongoing household surveys would enable dynamic assessment as new ground truth emerges. The Demographic and Health Surveys and Living Standards Measurement Studies provide repeated cross-sections that could support rolling validation, allowing performance assessment to keep pace with methodological updates and changing electrification contexts. Testing HREA's ability to detect electrification changes over time, not just levels, would assess the methodology's value for monitoring applications central to SDG 7 tracking. For development institutions, the ability to track progress reliably is as important as the ability to target investments accurately.

6.4 Closing

The High Resolution Electricity Access dataset represents a genuine advance in measuring global energy poverty. Its identification of 1.18 billion people in energy-poor settlements has reshaped understanding of the electrification challenge and attracted billions in development investment. This study confirms that HREA meets high standards for transparency and reproducibility. It also demonstrates that methods calibrated in one regional context do not automatically transfer to another. As satellite-based analytics become further embedded in development finance infrastructure, systematic validation across deployment contexts is not a luxury but a necessity. The tools exist to measure energy poverty from space. The task now is ensuring those measurements are reliable wherever they are applied.

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Notes

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